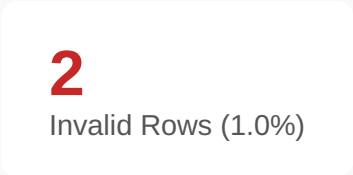
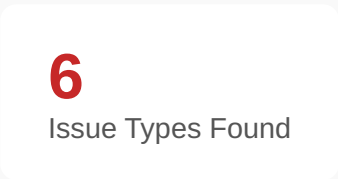
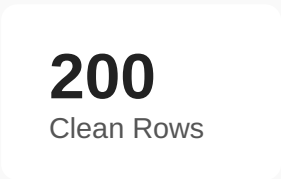
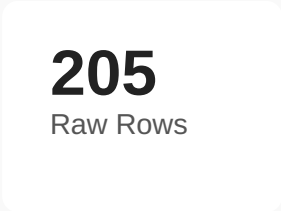
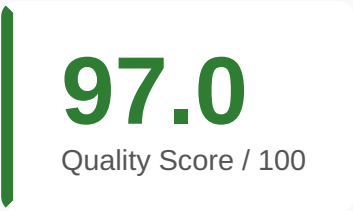


# AI Dataset Cleaning & Validation Report

Generated: 2026-05-13 22:05:48 | Validation: X FAILED



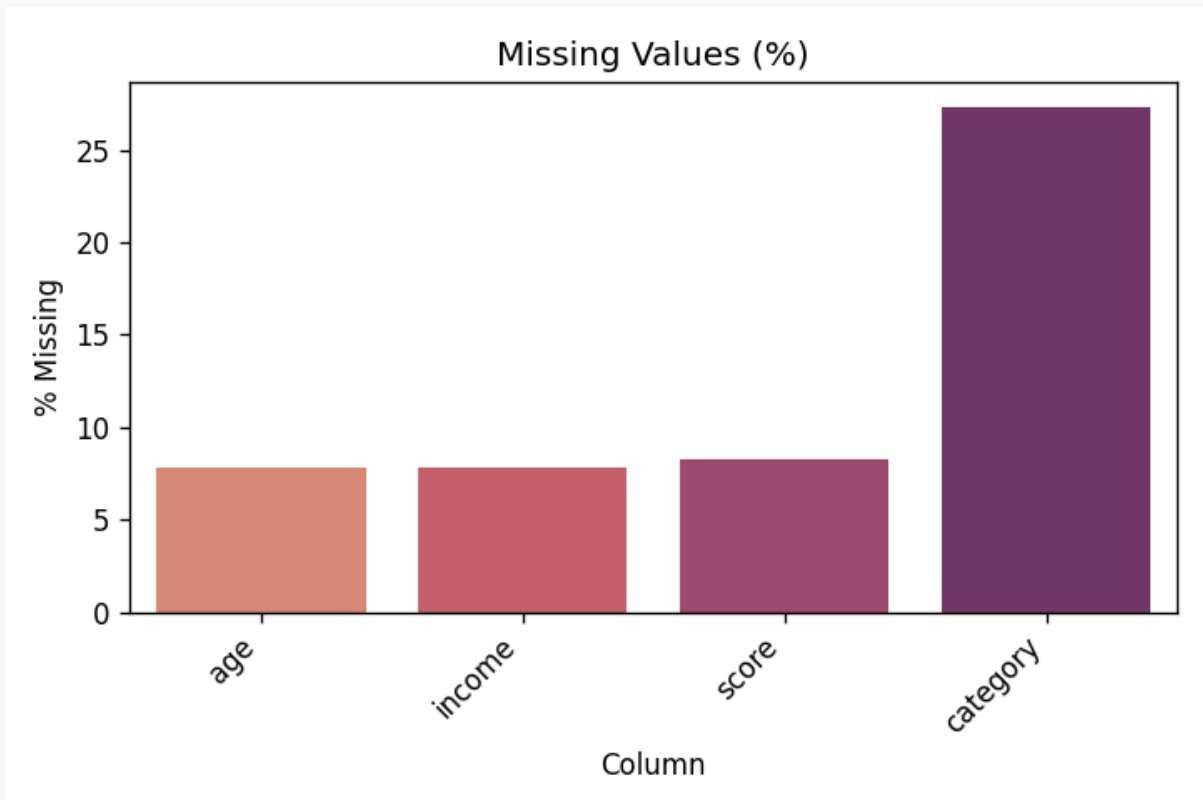
## 1. Dataset Overview

| Metric         | Raw      | After Cleaning |
|----------------|----------|----------------|
| Rows           | 205      | 200            |
| Columns        | 6        | 6              |
| Duplicate rows | 5        | 0              |
| Invalid rows   | 2 (1.0%) | —              |

## 2. Missing Values X FAILED

| Column | Missing Count | Missing % |
|--------|---------------|-----------|
| age    | 16            | 7.8%      |
| income | 16            | 7.8%      |
| score  | 17            | 8.29%     |

|          |    |        |
|----------|----|--------|
| category | 56 | 27.32% |
|----------|----|--------|



### 3. Column Profiles

| Column   | Dtype   | Null Count (%) | Unique | Min / Max / Mean / Median              |
|----------|---------|----------------|--------|----------------------------------------|
| id       | int64   | 0 (0.0%)       | 200    | 1 / 200 / 98.1 / 98                    |
| age      | float64 | 16 (7.8%)      | 55     | 18 / 130 / 46.8 / 46                   |
| income   | float64 | 16 (7.8%)      | 184    | 1.78e+04 / 1e+06 / 5.97e+04 / 4.82e+04 |
| score    | float64 | 17 (8.29%)     | 164    | 0.5 / 99.9 / 51.1 / 52.8               |
| category | object  | 56 (27.32%)    | 3      | —                                      |
| region   | object  | 0 (0.0%)       | 4      | —                                      |

### 4. Schema Violations ✓ PASSED

No schema violations detected.

## 5. Format Violations

✓ PASSED

All format checks passed.

## 6. Range Violations

✗ FAILED

| Column | Violations | Actual Range | Expected Range |
|--------|------------|--------------|----------------|
| age    | 2          | 18 – 130     | 0 – 120        |

## 7. Cleaning Log

- Standardized column names to snake\_case.
- Stripped whitespace from 2 string column(s).
- Removed 5 duplicate row(s).
- Filled 16 NaN(s) in 'age' with median (46).
- Filled 16 NaN(s) in 'income' with median (4.833e+04).
- Filled 16 NaN(s) in 'score' with median (54.2).
- Filled 53 NaN(s) in categorical 'category' with mode ('C').

## 8. Anomaly Detection (IQR)

| Column | Outlier Count | Outlier % |
|--------|---------------|-----------|
| age    | 1             | 0.5%      |
| income | 8             | 4.0%      |

## 9. Distributions (Clean Data)

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